

Hassan Muhammad Sharif

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Professional Summary

Software Engineering graduate specialising in AI/ML, with hands-on experience building RAG pipelines, autonomous AI agents, fine-tuned transformer models, and CNN-based systems. Proficient across the full ML lifecycle — from model training and vector search to REST API deployment. Passionate about building production-ready AI systems and solving real-world problems with modern ML engineering.

Technical Skills

AI / ML: PyTorch • LangChain • ChromaDB • HuggingFace • Scikit-learn • BERT • OpenCV

LLMs & RAG: LLaMA 3 • Groq • Ollama • Tavily • Prompt Engineering • Vector Search • RAGAS

Languages: Python • JavaScript • TypeScript • C++ • Ruby • HTML/CSS

Frameworks: FastAPI • Streamlit • React • Node.js • Django • Ruby on Rails

Tools: Docker • Git • Supabase • PostgreSQL • MongoDB • HuggingFace Hub

Projects

Research Agent — Autonomous AI Agent

github.com/hsnx999/research-agent

LangChain • Groq LLaMA 3.3 70B • Tavily Search • BeautifulSoup • Streamlit

- Built an autonomous ReAct agent that searches the web, scrapes pages, reasons across multiple steps, and saves structured markdown reports — without human intervention.
- Implemented 4 custom tools (web search, page scraping, report saving, calculator) with descriptive docstrings the LLM reads to decide which tool to invoke at each reasoning step.

DocChat — RAG-Powered Document Q&A Chatbot

[Live → Streamlit Cloud](#)

LangChain • ChromaDB • HuggingFace Embeddings • Groq LLaMA 3 • Streamlit • RAGAS

- Built a full RAG pipeline: PDF parsing, recursive chunking, vector embedding, ChromaDB storage, semantic retrieval, and streaming LLM response generation with conversation memory.
- Evaluated pipeline quality using RAGAS — achieving Faithfulness 0.95, Answer Relevancy 0.86, Context Precision 0.89 across 7 test cases.

Sentiment Analysis Pipeline — Fine-tuned BERT

github.com/hsnx999/sentiment-pipeline

HuggingFace Transformers • PyTorch • FastAPI • Streamlit • Plotly • SST-2

- Fine-tuned BERT-base-uncased on SST-2 achieving 87.25% accuracy; published model to HuggingFace Hub at hsnx999/bert-sentiment-sst2.
- Wrapped model in a FastAPI REST API (/predict and /predict/batch) and built a Streamlit dashboard with confidence bar charts and sentiment distribution visualisations.

Integrated AI Diagnostic Tool for Thoracic Oncology

FYP

React • FastAPI • CNNs • NLP • Supabase • Python

- Developed a full-stack AI diagnostic system using CNNs and NLP to analyze CT scans, automate TNM cancer staging, and generate predictive risk scores for lung cancer patients.
- Engineered a FastAPI backend for real-time CNN inference with Supabase integration and a React (Vite) frontend for clinical predictions.

Education

Bachelor of Science in Software Engineering

2021 – 2025

University of Lahore, Lahore

Experience

Ruby on Rails Intern — Inteldevs, Lahore

Jun – Sep 2024

- Contributed to production software module development and testing within an agile team; participated in code reviews and sprint planning.